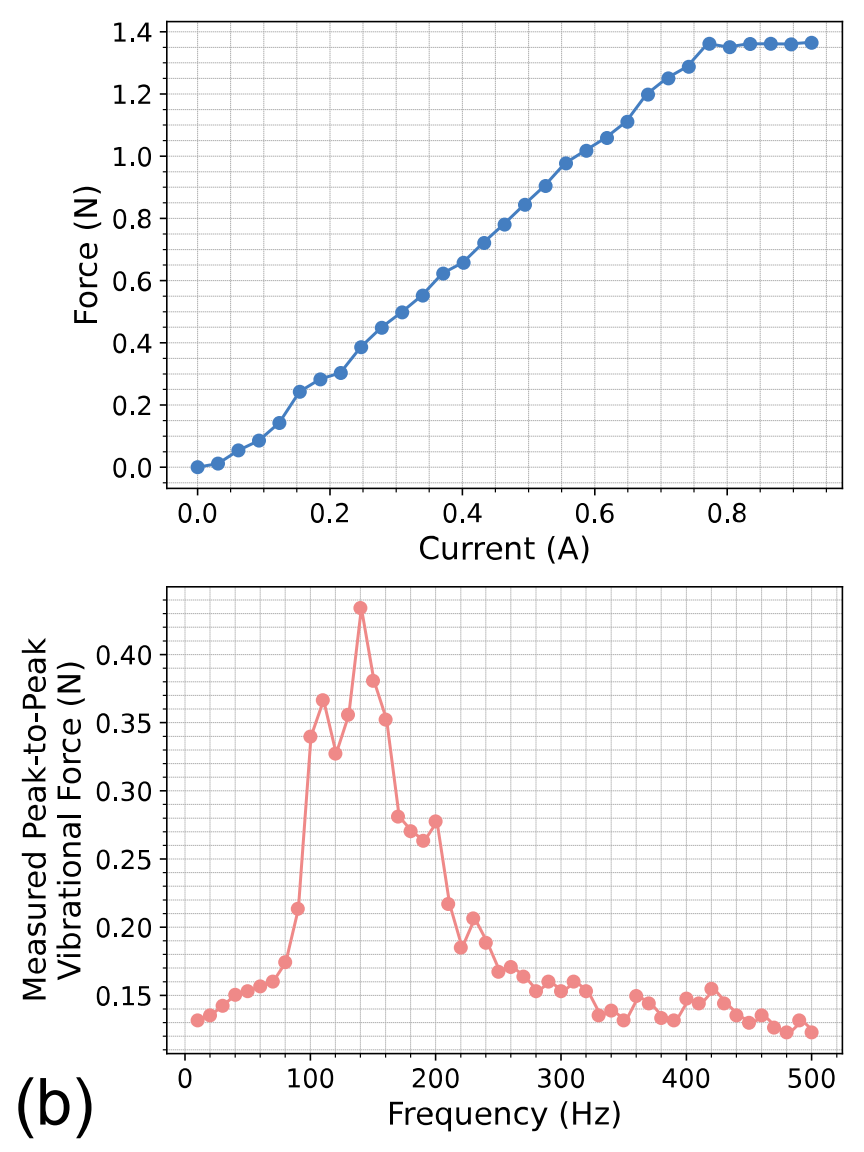

Force (N)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
Current (A)
Measured Peak-to-Peak Vibrational Force (N)
0.40
0.35
0.30
0.25
0.20
0.15
0
100
200
300
400
500
Frequency (Hz)
(b)